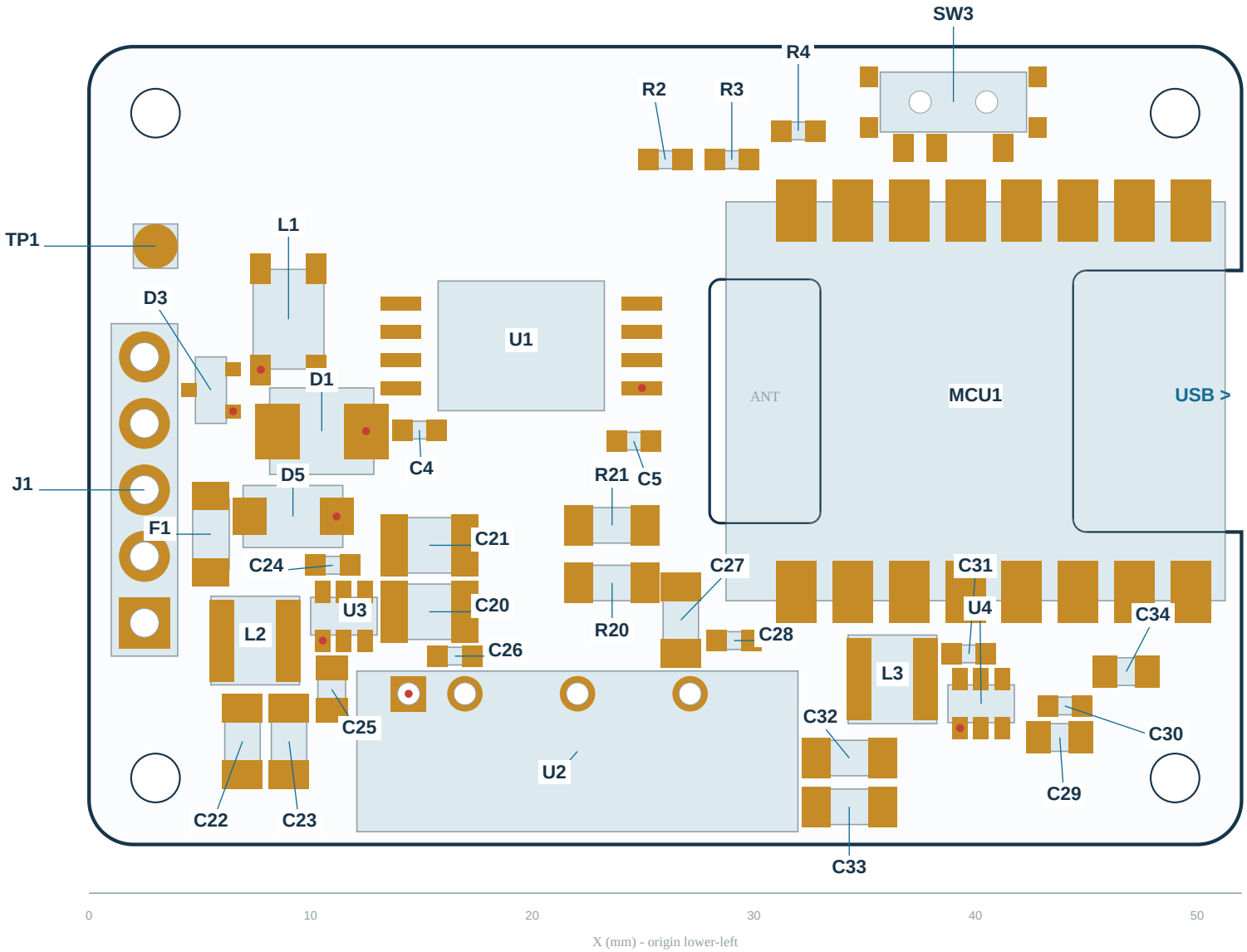
Hand assembly - Orion-R5

Release geometry, purchased components and a bench checklist | English documentation



Orientation: component side facing you; cable pads LEFT, USB notch RIGHT, service switch at TOP. Red dot = PCB pad 1 for directional parts. No bottom-side components.

Purchased parts - 1 or 4 boards

17 carrier groups / 34 carrier components per board, plus one module and one cable

References	Purchased MPN / LCSC	Package	1 / 4	Value and hand-assembly note
R2	0603WAF1002T5E C25804	0603	1 / 4	10 kOhm, 1%, 0.1 W
R3, R4	0603WAF330JT5E C23140	0603	2 / 8	33 Ohm, 1%
R20, R21	RNCP1206FTD150R C4258541	1206	2 / 8	150 Ohm, 1%, 0.5 W each; both required
C4, C5, C24, C26, C28, C30, C31	CC0603KRX7R9BB104 C14663	0603	7 / 28	100 nF, 50 V, X7R; non-polar ceramic
C20, C21	CL32B106KBJNNNE C138687	1210	2 / 8	10 uF, 50 V, X7R; non-polar ceramic
C22, C23, C27, C32, C33	CL31A226KAHNNNE C12891	1206	5 / 20	22 uF, 25 V, X5R; non-polar ceramic
C25, C29, C34	CL21A106KAYNNNE C15850	0805	3 / 12	10 uF, 25 V, X5R; non-polar ceramic
F1	SF-1206S080-2 C3032873	1206 fuse	1 / 4	0.8 A slow-blow, 63 VDC; non-polar
L2, L3	FNR4030S4R7MT C167874	4 x 4 mm	2 / 8	4.7 uH; purchased replacement; two-terminal, non-polar
D1	SMBJ18A C152104	SMB	1 / 4	Unidirectional TVS: cathode stripe toward RIGHT
D5	SS26 C14238	SMA	1 / 4	Schottky: cathode stripe toward RIGHT
D3	PESD2CAN24T-QR C28933048	SOT-23	1 / 4	Single lead LEFT; pin 1 lower-right, pin 2 upper-right
U3, U4	AP63205WU-7 C2071056	TSOT-26	2 / 8	Pin 1 lower-left in board top view; fixed 5 V version
U1	ISO1042DWVR C2671714	SOIC-8 DWV	1 / 4	Pin 1 lower-right in board top view
L1	PSCIAQ4532-510Z C2932173	4.5 x 3.2 mm	1 / 4	Purchased replacement; winding pairs 1-4 and 2-3
U2	B0505S-3WR2 C7500927	SIP-4 THT	1 / 4	Pin row LEFT to RIGHT: 1, 2, 4, 6; body toward lower edge
SW3	MK-12C02-G020 C963206	side SPDT	1 / 4	Actuator toward TOP board edge; verify RUN/SERVICE electrically

Additional: MCU1 = selected C3 SuperMini (1 / 4); J1 = NMEA cable (1 / 4); TP1 = bare copper, buy nothing. Quantities are exact board needs, with no loss/spare allowance. Package sizes 0603/0805/1206/1210 use imperial codes.

Per-reference placement and checklist

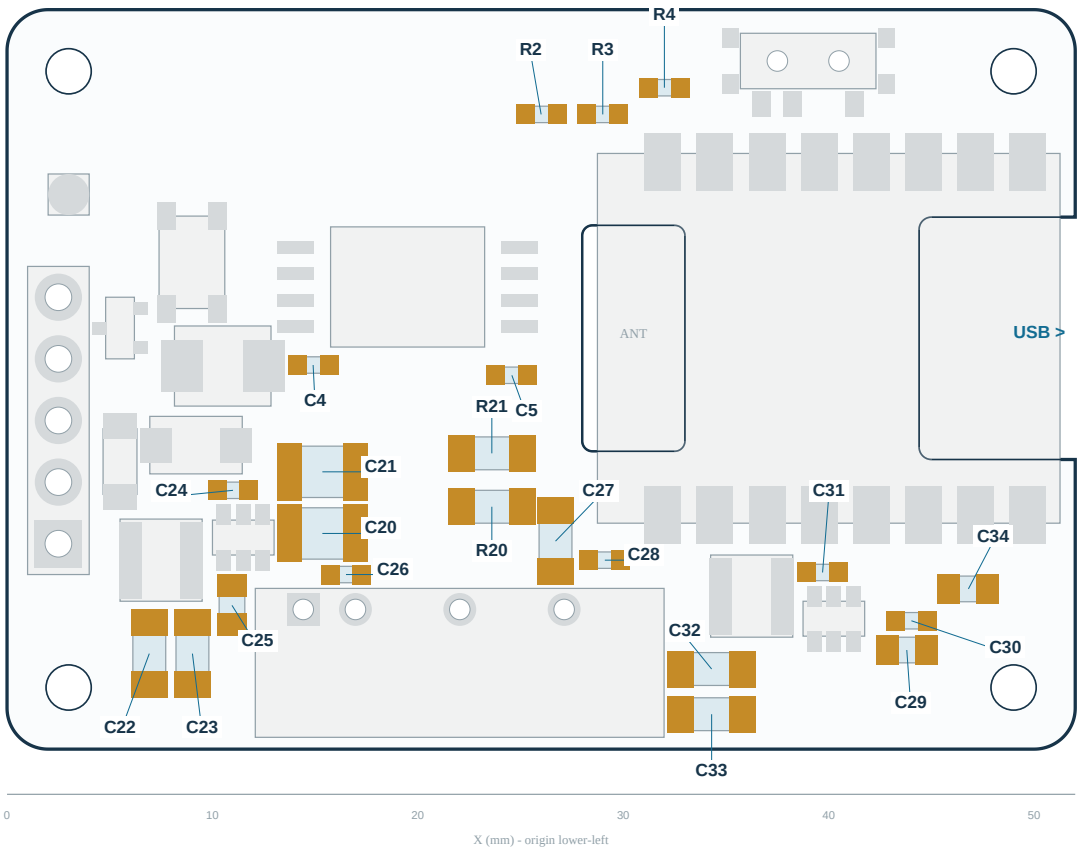
Mark the blank column after soldering and inspection; positions are CPL centres, in millimetres

Done	Ref	Release value	Purchased code	Centre X / Y	Rot.	Handling
	C4	100nF 50V X7R	C14663	14.900 / 18.700	0	Non-polar
	C5	100nF 50V X7R	C14663	24.575 / 18.200	0	Non-polar
	C20	10uF 50V X7R	C138687	15.350 / 10.500	0	Non-polar
	C21	10uF 50V X7R	C138687	15.350 / 13.500	0	Non-polar
	C22	22uF 25V X5R	C12891	6.925 / 4.650	270	Non-polar
	C23	22uF 25V X5R	C12891	9.025 / 4.650	270	Non-polar
	C24	100nF 50V X7R	C14663	11.000 / 12.600	0	Non-polar
	C25	10uF 25V X5R	C15850	10.950 / 7.000	90	Non-polar
	C26	100nF 50V X7R	C14663	16.500 / 8.500	0	Non-polar
	C27	22uF 25V X5R	C12891	26.700 / 10.125	90	Non-polar
	C28	100nF 50V X7R	C14663	29.100 / 9.200	0	Non-polar
	C29	10uF 25V X5R	C15850	43.800 / 4.825	0	Non-polar
	C30	100nF 50V X7R	C14663	44.025 / 6.250	0	Non-polar
	C31	100nF 50V X7R	C14663	39.700 / 8.600	0	Non-polar
	C32	22uF 25V X5R	C12891	34.300 / 3.900	0	Non-polar
	C33	22uF 25V X5R	C12891	34.300 / 1.700	0	Non-polar
	C34	10uF 25V X5R	C15850	46.800 / 7.800	0	Non-polar
	D1	SMBJ18A 600W	C152104	10.500 / 18.650	180	See orientation details
	D3	PESD2CAN24T-Q	C28933048	5.500 / 20.500	90	See orientation details
	D5	SS26 2A 60V	C14238	9.200 / 14.800	180	See orientation details
	F1	T0.8A 63VDC	C3032873	5.500 / 14.000	90	Non-polar
	J1	NMEA Micro-C pigtail	Amazon	2.500 / 16.000	180	Wire termination
	L1	ACT45B-510-2P-TL003	C2932173	9.000 / 23.700	90	See orientation details
	L2	4.7uH	C167874	7.500 / 9.200	180	Non-polar
	L3	4.7uH	C167874	36.250 / 7.450	180	Non-polar
	MCU1	ESP32-C3 SuperMini	Amazon	40.811 / 20.000	270	See orientation details
	R2	10k 1% 0.1W	C25804	26.000 / 30.900	0	Non-polar
	R3	33R	C23140	29.000 / 30.900	0	Non-polar
	R4	33R	C23140	32.000 / 32.200	0	Non-polar
	R20	150R 1% 0.5W	C4258541	23.600 / 11.800	0	Non-polar
	R21	150R 1% 0.5W	C4258541	23.600 / 14.400	0	Non-polar
	SW3	RUN / SERVICE	C963206	39.000 / 33.030	0	See orientation details
	TP1	Shield / chassis pad	-	3.000 / 27.000	0	Bare pad; no part
	U1	ISO1042DWV	C2671714	19.500 / 22.500	180	See orientation details
	U2	B0505S-3WR2	C7500927	20.770 / 6.800	0	See orientation details
	U3	AP63205WU-7	C2071056	11.500 / 10.300	0	See orientation details
	U4	AP63205WU-7	C2071056	40.250 / 6.350	0	See orientation details

Angles are the exported Altium rotations, not universal machine-rotation conventions. Use the top-view diagrams and labelled pads for manual orientation. The CSV also records component origins separately from geometric centres.

Step 1 - resistors and ceramic capacitors

Sort parts into labelled containers; solder low-profile parts before tall modules



Resistors
R2: 10 kOhm. R3/R4: 33 Ohm. R20/R21: 150 Ohm, 0.5 W EACH. Fit both preload resistors.

100 nF / 0603
C4, C5, C24, C26, C28, C30, C31: 50 V X7R. All seven use the same purchased part.

10 uF / 1210
C20, C21: 50 V X7R. Keep these apart from the smaller 25 V capacitors.

22 uF / 1206
C22, C23, C27, C32, C33: 25 V X5R.

10 uF / 0805
C25, C29, C34: 25 V X5R. C34 is behind the service switch.

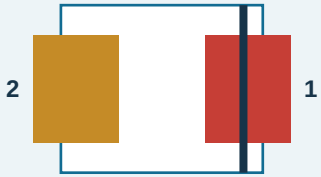
All listed capacitors are non-polar MLCCs. Use package labels to identify unmarked capacitors. Tack one end on a lightly tinned pad, align, solder the other end, then inspect both joints. Avoid prolonged heating and mechanical stress on ceramic parts.

Before the next step: inspect for bridged pads and wrong values under magnification. The release has no C1-C3, C6-C19, R1, D2 or D4 carrier positions; do not add parts to nonexistent references.

Step 2 - diodes and buck regulators

TOP VIEW | Both diode cathode stripes point RIGHT, toward the module side

D1 - SMBJ18A



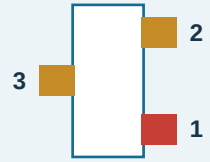
1 = K / BUS_PROT (RIGHT)
2 = A / GND_BUS (LEFT)

D5 - SS26



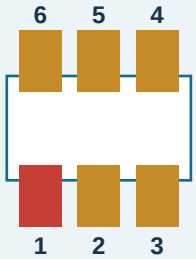
1 = K / BUS_PROT (RIGHT)
2 = A / FUSED_IN (LEFT)

D3 - PESD2CAN24T-QR



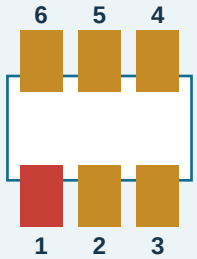
1 = CANH_EXT, lower-right
2 = CANL_EXT, upper-right
3 = GND_BUS, single lead LEFT

U3 - AP63205WU-7



1 FB / 2 EN / 3 VIN below
6 BST / 5 SW / 4 GND above

U4 - AP63205WU-7



Same pin orientation as U3.
Use AP63205, fixed 5 V.

Regulator joints

Identify physical pin 1 from the component package marking. Match the numbered PCB pad; tack opposite corners, inspect alignment, then solder the remaining leads with flux.

Check all six leads for bridges. U3 uses GND_BUS; U4 uses GND_LOGIC. Preserve the isolation gap.

Step 3 - isolated CAN and inductors

TOP VIEW | Fit F1, L1, L2 and L3 after checking part identity and winding orientation

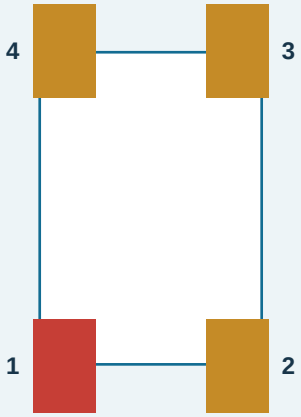
U1 - ISO1042DWVR



Pin 1 is LOWER-RIGHT. Locate the package pin-1 marker.
Logic leads RIGHT; bus leads LEFT.

U1 lead	Function	PCB location
1 / 2 / 3 / 4	VCC1 / TXD / RXD / GND1	RIGHT, bottom to top
8 / 7 / 6 / 5	VCC2 / CANH / CANL / GND2	LEFT, bottom to top

L1 - PSCIAQ4532-510Z



L1 replacement: PSCIAQ4532-510Z / C2932173.
PCB winding paths must run bottom-to-top, not across.

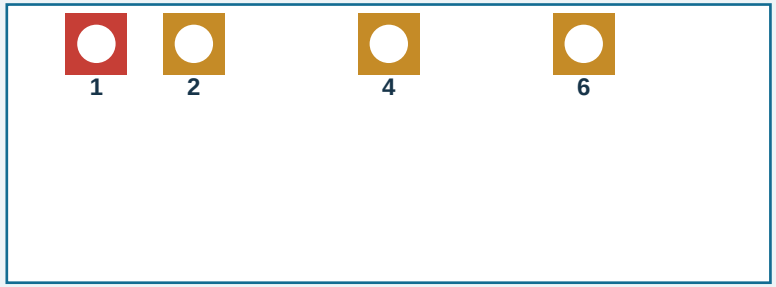
L1: pad 1 lower-left = CANH_EXT; 4 upper-left = CANH. Pad 2 lower-right = CANL_EXT; 3 upper-right = CANL. Before soldering, use continuity to identify the two separate windings (1-4 and 2-3). The replacement sheet does not provide a clear external pin-1 mark; do not infer one from this drawing.

F1: SF-1206S080-2, non-polar. **L2/L3:** FNR4030S4R7MT, 4.7 uH, non-polar two-terminal inductors. L1 has four terminals and is a different component. Replacement fit was reviewed digitally; check wetting of all four L1 terminals on the actual PCB.

Step 4 - converter and service switch

TOP VIEW | Install tall U2 after the nearby small components; keep switch travel clear

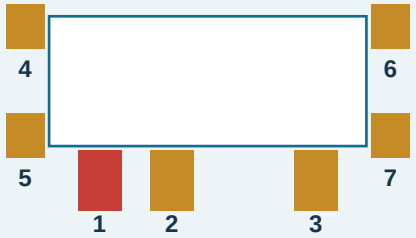
U2 - B0505S-3WR2



Four THT leads: 1, 2, 4, 6 LEFT to RIGHT.
Row at Y=6.80 mm; body lies mainly BELOW that row.

U2 lead	Net	Role
1	5V_BUS	+ input
2	GND_BUS	Input ground
4	GND_LOGIC	Isolated 0 V
6	5V_ISO	Isolated + output

SW3 - MK-12C02-G020



Contacts 1, 2, 3; shell pads 4-7.
The two locating holes remain free of solder.

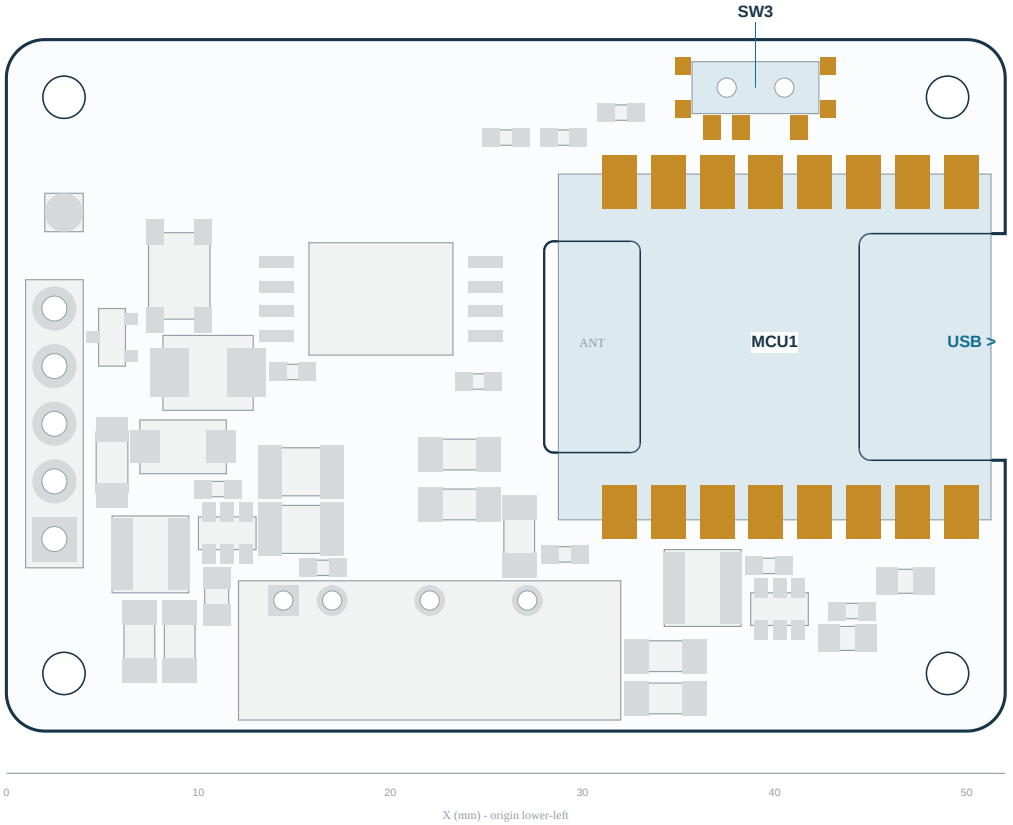
SW3 electrical check, power OFF
RUN: contact 1 (5V_REG) connects to contact 2 (5V_MODULE). SERVICE: contact 2 connects to unconnected contact 3; 1-2 is open. Shell tabs 4-7 connect only to GND_LOGIC.

The release silkscreen is a location aid. Confirm actual lever positions by continuity; the supplier drawing does not establish left/right travel.

USB sequence: select SERVICE before plugging in USB. Remove USB before returning to RUN. Do not bridge SW3, and do not connect bus ground to logic ground.

Step 5 - ESP32-C3 SuperMini

Selected Amazon C3 module, ASIN B0FN4F6244 | USB RIGHT, antenna LEFT



Dry fit before soldering
Match all 16 castellations to their carrier pads. Remove any headers and solder protrusions. Check the USB underside over the right-edge notch and the antenna over the internal window.

Carrier pads are for direct perimeter soldering. Use no carrier headers, no paste under the module and no conductive adhesive. Tack opposing corners, recheck fit, then solder all 16 lands, including unused pins.

X (mm)	Upper row Y=28.60	Lower row Y=11.40
31.921	GPIO21 / unused	GPIO0 / unused
34.461	GPIO20 / unused	GPIO1 / unused
37.001	GPIO10 / unused	GPIO2 / unused
39.541	GPIO9 / unused	GPIO3 / unused
42.081	GPIO8 / unused	GPIO4 / unused
44.621	GPIO7 / CAN RX	3V3 / 3V3_LOGIC
47.161	GPIO6 / CAN TX	GND / GND_LOGIC
49.701	GPIO5 / unused	5V / 5V_MODULE

GPIO6 = CAN TX; GPIO7 = CAN RX. The module 3V3 pin is an OUTPUT to U1/C5. Verify the delivered C3 pin labels; reject a different pinout. Mechanical clone fit and radio operation remain physical acceptance checks.

Step 6 - cable termination and inspection

J1 is five solder holes, not a purchased board connector | Continuity-check the actual cable

J1 pad	Y (mm)	Signal	Expected cable conductor	Connection
5 (TOP)	22.00	CAN-L	Blue	CANL_EXT
4	19.00	CAN-H	White	CANH_EXT
3	16.00	NET-C	Black	GND_BUS
2	13.00	NET-S	Red	Bus supply input
1 (BOTTOM, square)	10.00	SHIELD	Bare drain / shield	SHIELD; also accessible at TP1

All five hole centres are at X=2.50 mm. Wire colours are the purchased pigtail listing convention, not evidence for the delivered cable. Check connector-contact-to-wire continuity before soldering. This table is a PCB top view; do not use it as a Micro-C connector face view.

- [ ] Thread the unsplit cable seal/cartridge and cable-entry parts onto the cable BEFORE soldering. Check stripped/tinned conductors fit the 1.3 mm holes without force.
- [ ] Keep shield strands from all other pads. SHIELD, GND_BUS and GND_LOGIC are separate nets; add no ground or shield jumpers.
- [ ] Provide enclosure strain relief; solder joints must not carry cable pull. Trim excess wire and THT lead ends so they do not short to the enclosure or adjacent copper.
- [ ] Inspect every joint, module land, U1 lead, U3/U4 lead and L1 terminal. Remove solder bridges and clean residues according to the flux instructions.
- [ ] With all power and USB disconnected: check input/rail shorts, polarity and RUN/SERVICE continuity. Capacitor charging can produce transient meter readings; investigate stable low resistance.

First power-up - record the actual results

Use an isolated bench setup before connecting to the installed NMEA network

- ☐ Begin with MCU1 not fitted if practical: after carrier completion, select SERVICE, leave USB disconnected and test the carrier supplies with a current-limited source at J1 NET-S/NET-C. Use the documented 9-16 V bus range; stop for excessive current, smoke or fast heating.
- ☐ Measure 5V_BUS relative to GND_BUS at U2 pin 1. Measure 5V_ISO relative to GND_LOGIC at U2 pin 6. Measure 5V_REG relative to GND_LOGIC at SW3 pin 1. Do not measure an isolated rail against bus ground.
- ☐ U2 is unregulated. Its output is not guaranteed to be exactly 5.00 V; compare measurements to the power review. U4 can enter dropout and cannot boost a low U2 output. Verify startup and unloading peaks before attaching the module.
- ☐ Remove bench power before fitting MCU1. For USB-only service, select SERVICE first; verify module boot and 3V3_LOGIC. Disconnect USB before switching to RUN and testing bus-powered operation.
- ☐ Verify no physical bridge connects GND_BUS to GND_LOGIC or SHIELD. Use a multimeter for accidental-short checks; do not apply a high-voltage insulation tester to the assembled electronics.
- ☐ When using an earth-referenced oscilloscope, do not attach grounded clips to both isolated ground domains. Use an appropriate differential/isolated measurement arrangement.
- ☐ Record bus current, all rails, switching transitions and temperatures. Validate BLE reception, service Wi-Fi, CAN/NMEA traffic and Garmin values separately. Local firmware counters do not establish receiver delivery.

Board serial: _____ Assembler: _____ Date: _____

Input V / A: _____ 5V_BUS: _____ 5V_ISO: _____ 5V_REG: _____

Module 5V / 3V3: _____ Startup peak: _____ Thermal / communication result: _____

Sources, substitutions and remaining checks

Release files remain unchanged; this package is an assembly aid, not new hardware qualification

Release authority

hardware/altium-r5/Project Outputs/BatterySentinel-N2K-Orion-R5.zip

ASCII PCB and Pick Place export: 23 September 2026, 20:53. Placement, pad numbers, pad coordinates, footprint outlines and nets are read directly from that archive. The native BOM provides release values and reference coverage.

Purchased replacements

L1: PSCIAQ4532-510Z / C2932173 replaces the release ACT45B-510-2P-TL003. L2/L3: FNR4030S4R7MT / C167874 replace the planned Sunlord part; the SUNLORD_SWPA4030 footprint remains. These changes affect only the hand-assembly BOM. Local procurement records dated 23/24 September document the review; supplied parts still need identity and fit inspection.

Local engineering references

lcsc-order-2026-09-24.md; lcsc-preisvergleich-2026-09-23.md; power-review.md; module-selection.md.

Datasheets retained in source-references: PROD L1, cjiang L2/L3, EVISUN converter, Diodes buck regulator, G-Switch. Original release schematic and assembly PDFs are included for cross-checking.

Known documentation distinctions

The release generic C4/C5 description mentions 16 V, but their release value and purchased CC0603KRX7R9BB104 are 100 nF / 50 V. Use the value and purchased MPN. CPL geometric centres differ from component origins for U2, SW3 and MCU1; both are labelled separately in the position CSV. All geometry comes from the final release, avoiding later development-file placement assumptions.

Completed digital checks

37 references agree across release BOM, CPL and ASCII PCB; 34 purchased carrier components plus module, cable and bare pad; all 114 numbered pad nets checked against design connectivity. The existing Altium DRC report is not a newly executed DRC.

Open physical work: component identity and fit, soldering, short/polarity checks, switch continuity, rail/startup/thermal measurements, RF/CAN/Garmin acceptance and enclosure retention/sealing.